

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

*I Aaron Samuel S (192411022) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. SOAP stands for Simple Object Access Protocol. It is a protocol used for exchanging structured information in web services.
- 2.WSDL stands for Web Services Description Language. It describes the operations, messages, and location of a web service.
- 3.An API (Application Programming Interface) allows different software applications to communicate. It defines rules and methods for accessing data or services.

- 4.XML Schema defines the structure and data types of an XML document. It specifies the elements and attributes allowed in the document.
- 5.A Web Service is a software system that enables communication between applications over a network. It allows different platforms and applications to exchange data.
- 6.REST stands for Representational State Transfer. It is an architectural style for building web services using HTTP methods and resources.
- 7.A protocol is a set of rules that defines how data is communicated between systems. It ensures that devices and applications can exchange information correctly.
- 8.An endpoint is a specific URL where a web service or API can be accessed. It is the location where a client sends requests to a service.
- 9.A service is a software function that performs a specific task for an application. It provides data or functionality to other applications through a network.
- 10.SOAP is a protocol that commonly uses XML and follows strict standards. REST is an architectural style that commonly uses HTTP and formats such as JSON.
- 11.WSDL describes what a web service does and how it can be accessed. It helps clients understand the available operations, messages, and service location.
- 12.JSON (JavaScript Object Notation) is a lightweight format for representing data. It is widely used in APIs to send and receive data between clients and servers.
- 13.HTTP (HyperText Transfer Protocol) is used to transfer data over the web. Web services use HTTP to send requests and receive responses between clients and servers.
- 14.A request is a message sent by a client asking a server to perform an operation. A response is the server's reply containing the requested data or result.
- 15.XML is used to represent and exchange structured data in web services. It is widely used in SOAP messages and for defining data structures.
- 16.An API endpoint provides a specific location for accessing a particular resource or operation. Clients use endpoints to send requests and receive responses from an API.
- 17.HTTP transfers data without encryption, while HTTPS encrypts the communication. HTTPS provides greater security by protecting data from being intercepted.
- 18.RESTful services are web services designed according to REST principles. They use resources, URLs, and HTTP methods such as GET, POST, PUT, and DELETE.
- 19.A namespace in XML uniquely identifies XML elements and prevents naming conflicts. It allows elements with the same name to be distinguished from different vocabularies.

20. The client sends requests to access a web service or resource. The server receives the requests, processes them, and sends back responses.

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